SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Project Initialization and Planning Phase

Date	30 june 2026
Team ID	
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform credit card approval screening using machine learning, boosting efficiency and accuracy. It tackles the inefficiencies of manual application review, promising faster operations, reduced risk exposure, and a better experience for applicants. Key features include a machine learning-based credit approval model and real-time, cloud-hosted decision-making.

Project Overview	
Objective	The primary objective is to automate the credit card approval process by implementing machine learning techniques, ensuring faster and more accurate approve/reject decisions.
Scope	The project comprehensively assesses applicant financial and demographic data to predict credit card approval outcomes, incorporating machine learning for a more robust, consistent, and efficient screening system that supports analysts, compliance staff, and applicants.
Problem Statement	
Description	Banks receive thousands of credit card applications daily, and manually reviewing each one against factors such as loan balance, income level, and credit inquiries is time-consuming and highly prone to human error, leading to inefficiency and inconsistency at scale.
Impact	Solving these issues will result in faster application turnaround, more consistent and accurate approval decisions, reduced risk from missed high-risk applicants, and an overall improvement in applicant satisfaction and organizational efficiency.
Proposed Solution	
Approach	Employing machine learning techniques to analyze historical applicant data and predict creditworthiness, creating a dynamic and adaptable credit card approval system accessible through a web application.

Key Features	- Implementation of a machine learning-based credit approval model, benchmarking Logistic Regression, Random Forest, XGBoost (Gradient Boosting), and Decision Tree, with the best-performing model selected for deployment. - Real-time decision-making for instant approval or rejection predictions via a Flask web application. - Feature engineering pipeline that converts multi-class payment-status codes into binary risk labels for clearer high-risk applicant identification. - Cloud deployment through an IBM Watson Machine Learning pipeline, enabling scalable, real-time predictions accessible through an intuitive user interface. - Supports multiple usage scenarios: analyst-driven screening, compliance batch review of high-risk applicants, and customer self-service eligibility checks.

Resource Requirements:

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	16 GB
Storage	Disk space for data, models, and logs	512 GB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, xgboost, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, VS Code

Cloud Deployment	Cloud ML hosting platform	IBM Watson Machine Learning
Data		
Data	Source, size, format	Credit card approval dataset (UCI / Kaggle), ~690 records, CSV